Food Waste Reduction Mobile Application, A Rigorous System Analysis

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***Abstract*—In Colombia, fruits and vegetables represent 62% of food waste (6.1 million tons), while roots and tubers account for 25% (2.4 million tons). Amid the global crisis of food loss and waste—which exacerbates environmental damage and insecurity—this underscores the urgent need for collaborative efforts by NGOs, foundations, businesses, and individuals. Digital platforms serve as essential enablers by reducing coordination barriers among actors, streamlining surplus food collection and redistribution processes, and primarily facilitating real-time connections between donors (such as university cafeterias and restaurants) and recipients (charities, students, and communities), thus contributing to measurable reductions in waste, enhanced social equity, and alignment with circular economy principles.**

***Index Terms*—Food Waste, Surplus food redistribution, Geolocation**

I. System Overview and Scope Definition

A. Comprehensive System Description

The Food Waste Reduction Platform is a system designed to connect university cafeterias and local restaurants with students and the community in Bogotá, facilitating the real-time redistribution of surplus food that remains suitable for human consumption. The system integrates a mobile application (used for coordination, logistics management, and delivery tracking) with a series of operational processes and strategic partnerships among actors in the university and commercial environment. Its primary objective is to reduce food waste by creating an efficient redistribution channel, enabling establishments that generate surplus to offer it quickly and organized to nearby users interested in accessing it.

1) General System Context: The system emerges as a response to the problem of food waste in urban environments. In Bogotá, university cafeterias and restaurants generate significant amounts of surplus food daily. The platform seeks to intervene in this context through a digital coordination mechanism that facilitates the rapid redistribution of these foods before they lose their utility.

B. System Boundaries

For the purposes of this analysis, the system is delimited by a set of functional, spatial, temporal, and technological boundaries that clearly establish which elements are part of the system and which belong to its environment.

1) Within the System (Controlled and Managed by the Platform):

- Registration and management of users, including students, foundations, cafeterias, and partnered restaurants.
- Geolocation of pickup and delivery points within an approximate radius of 3 km from the Engineering Faculty headquarters of the Universidad Distrital Francisco José de Caldas.
- Coordination and matching between suppliers and demanders through a proximity- and availability-based matching algorithm.
- Historical record of collectors and tracking of redistributions.
- Metrics such as the amount of food recovered and estimates of waste reduction.

2) Outside the System (Not Controlled by the Platform):

- Physical transportation of food from the establishment to the point of consumption (responsibility of the user, volunteer, or interested entity).
- Physical verification of the quality, condition, or food safety of the offered foods.
- Payment or economic compensation processes between parties, which are not part of the initial scope of the system.
- Internal inventory management, preparation, or storage of food within each participating cafeteria, restaurant, or foundation.

C. Spatial Boundaries

The system is initially implemented in the vicinity of the Engineering Faculty headquarters of the Universidad Distrital Francisco José de Caldas (located at Carrera 8 # 40-62, Calle 40, Bogotá). These delimitations facilitate logistical coordination of redistributions.

- **Central Point:** Engineering Faculty headquarters of the Universidad Distrital Francisco José de Caldas.
- **Maximum Operating Radius:** Approximately 3 km from the central point, ensuring primarily pedestrian accessibility or short trips.

- **Exclusions:** Formal food bank donations outside this radius and surplus from chain restaurants located in more distant areas are not considered within the system's scope.

D. Temporal Boundaries

1) System Operating Hours: The system operates within a time window aligned with the usual operating hours of cafeterias and restaurants.

- **Operating Hours:** From 7:00 a.m. to 9:00 p.m., every day of the week.
- **Surplus Lifecycle:** A surplus can be published on the platform approximately 2 to 4 hours before the establishment closes, with an estimated collection window of 1 to 3 hours.
- **Exclusions:** Management of surplus outside the same day of publication is not contemplated, in order to guarantee the freshness and safety of the foods.

E. Main Components

1) Critical System Components: The Food Waste Reduction Platform comprises several critical components that enable its core functionality: real-time surplus food redistribution with priority to verified organizations. These components are grouped by functional category and detailed below, including their scope (within or outside the controlled system), rationale for criticality, and key dependencies/interactions. A complete breakdown of all components, including detailed descriptions and dependency mappings, is available in the project documentation (see `Workspace/docs/system_components.xlsx`).

Category	Component
User Interface	Mobile Application (Frontend)
Backend & Persistence	Cloud Server + Database
Intelligence Engine	Matching Algorithm

F. Inputs, Processes, and Outputs

This subsection describes the primary inputs to the Food Waste Reduction Platform, the main processes that transform these inputs into value, and the key outputs delivered by the system. The design follows a classic input-process-output (IPO) model, ensuring traceability from data entry to measurable impact.

1) Inputs: The system receives the following key inputs from stakeholders and automated sources:

- **Descriptive Information of Surplus Food Items:** Provided by suppliers (cafeterias or restaurants) at the time of registering an available surplus. This includes details such as type of food, quantity (in kg or portions), estimated expiration time, photos, and any relevant notes (e.g., allergens or preparation status).
- **Geospatial Data of Pickup Points:** Automatically generated by the supplier's mobile device during surplus publication. This includes precise or approximate GPS coordinates, address, and operational radius constraints.
- **User Profiles and Preferences of Demanders:** Collected from registered recipients (students, community members, charities/foundations). Includes basic food preferences, dietary restrictions, consumption patterns, and location data (for proximity filtering).
- **Real-Time Collection Requests:** Generated by demanders upon viewing a proposed surplus item. These include explicit interest signals, requested pickup time windows, and any additional comments.

2) Main Processes: The core value chain of the platform is driven by the following sequential and automated processes:

- **Surplus Publication by the Supplier:** The supplier registers a surplus item through the mobile application. The system performs basic validation (completeness, expiration validity, location accuracy) and immediately activates the matching process.
- **Automatic Surplus Distribution and Matching:** Real-time evaluation of registered demanders based on proximity (distance), remaining shelf life, priority rules (verified charities first), and basic preference alignment. The algorithm generates a ranked list of candidates and selects the best match according to predefined criteria.
- **Notification and Acceptance by the Selected Demander:** The system sends a real-time notification (push or in-app) to the selected user. The recipient has a limited time window to accept or reject the offer. If rejected, the system re-triggers matching to the next candidate.
- **Tracking and Closure of the Collection:** Upon acceptance and confirmation of pickup by both parties (supplier confirmation + demander proof-of-collection, e.g., photo or QR scan), the system logs the successful redistribution and monitors completion status.
- **Automatic Update of Impact Metrics:** At the closure of each successful collection, the system automatically aggregates and updates key performance indicators (e.g., total kg recovered, number of transactions, priority fulfillment rate for charities).

3) Outputs: The platform generates the following primary outputs, which represent the tangible and measurable results of its operation:

- **Redistributed and Utilized Food:** Successful delivery of surplus food that would otherwise have been wasted, directly contributing to food security for recipients and waste prevention.
- **Impact Data and Performance Indicators:** Quantitative metrics such as total kilograms of food recovered, number of successful transactions per period, success rate by priority tier (charities vs. individuals), estimated environmental savings (e.g., CO_2 equivalent avoided), and user engagement statistics.
- **Confirmation Notifications:** Final confirmation messages sent to both the supplier (successful pickup) and the demander (receipt acknowledgment), closing the transaction loop and enabling feedback collection.

G. Stakeholder Summary

The following is a summary of the main stakeholders involved in the Food Waste Reduction Platform, their roles, importance within the system, and how they are represented in the platform architecture.

1) Users: The following stakeholders interact directly with the platform as end users:

- **Students:** Primary consumers of surplus food and key drivers of platform demand. Registered mobile application users who can browse, reserve, and receive notifications about available food.
- **Local Community Members:** Increase the likelihood that surplus food will be claimed and redistributed efficiently. Verified user accounts with access to food listings and reservation features.
- **Charities and Foundations:** Enable redistribution of larger quantities of food to vulnerable populations. Verified institutional accounts with potential priority access to surplus listings.
- **University Cafeterias:** Stable potential source of surplus food within university environments. Vendor accounts that can publish surplus food listings and manage availability.
- **Partner Restaurants:** Increase the diversity and availability of food within the platform ecosystem. Verified vendor accounts capable of creating and managing surplus food listings.

2) Operators: The following stakeholders are responsible for maintaining the platform's operational continuity:

- **Technical Support Team:** Maintain system reliability and resolve technical incidents. Internal access to system monitoring tools, logs, and technical maintenance interfaces.
- **Operational Monitoring Staff:** Ensure proper interactions between participants and manage disputes or irregular activity. Moderation dashboards with access to activity reports and alerts.

3) Administrators: The following stakeholders oversee governance, management, and strategic development:

- **Platform Administrators:** Manage platform configuration and approve participating vendors. Administrative dashboards for user management and system configuration.
- **System Administrators:** Maintain infrastructure, security, and database integrity. Backend-level access to servers, databases, and configuration tools.
- **Project Management Team:** Define strategic direction, policies, and long-term system development. Access to analytics dashboards, system reports, and operational metrics.

4) External Actors: The following stakeholders influence the system indirectly without being direct users of the platform:

- **University Institutions:** Provide the institutional context where the system may be deployed. Represented through partnerships, governance agreements, and operational policies.
- **Public Health Authorities:** Define food safety regulations affecting redistribution practices. Represented through regulatory constraints implemented in the platform rules.
- **Cloud Infrastructure Providers:** Provide scalable infrastructure and ensure platform availability. Backend hosting services including servers, storage, and cloud databases.
- **Mobile Network Providers:** Enable real-time connectivity between users and the platform. External communication infrastructure supporting mobile network access.
- **Sustainability Organizations:** Support environmental impact initiatives and awareness programs. External stakeholders interacting through partnerships and impact reports.

H. Operating Environment Characterization

The operational environment of the Food Waste Reduction Platform is deliberately constrained to ensure feasibility, logistical efficiency, and food safety in the context of a university-led prototype project in Bogotá.

1) Physical and Spatial Environment: The system operates within a clearly defined geographic perimeter centered on the Engineering Faculty headquarters of the Universidad Distrital Francisco José de Caldas, located at Carrera 7 # 40B-53, Bogotá, Colombia. The maximum operational radius is set to approximately 3 km around this central point.

This limited spatial scope ensures that:

- All pickup points (participating university cafeterias and nearby local restaurants) remain within short travel distances.
- Collections can be performed primarily on foot, by bicycle, or via short public transport trips, minimizing logistical barriers for students, community members, and volunteers.
- Response times remain practical and aligned with the short shelf-life of perishable surplus food.

The choice of this 3 km radius corresponds to a realistic pedestrian-accessible zone in an urban university setting and facilitates initial pilot testing and partner recruitment in the immediate vicinity.

2) Temporal Environment: The platform functions within a strict daily operational window aligned with typical surplus generation patterns in academic and commercial food services.

- **Daily Operating Hours:** From 7:00 a.m. to 9:00 p.m., seven days a week. This interval covers the main meal periods (breakfast, lunch, and dinner) when surplus is most likely to occur in cafeterias and restaurants.
- **Surplus Lifecycle Constraints:**
  - Surplus items may be published on the platform approximately 2 to 4 hours before the establishment's closing time.

  - The maximum collection window is limited to 1–3 hours after publication to guarantee freshness, food safety, and compliance with basic hygiene standards.
- **No Overnight or Multi-Day Management:** The system does not support the accumulation or deferred management of surplus from one day to the next. Any unclaimed items at the end of the daily window are considered expired within the platform and are not carried over. This temporal restriction is imposed to:
  - Minimize health and safety risks associated with perishable foods.
  - Maintain high user trust by ensuring only fresh, same-day surplus is offered.
  - Simplify operational complexity during the prototype phase.

These temporal boundaries impose significant constraints on real-time coordination and matching algorithms but are essential to align the system with real-world food handling practices and to prevent misuse or degradation of redistributed items.

II. Data Collection Plan and Execution

The data collection for this project is framed within two distinct approaches:

- Collection from reliable and up-to-date secondary sources, to establish the context of the food waste problem in Colombia and Bogotá.
- Primary metrics generated directly by the platform prototype (logs of postings, matches, claims, and confirmed pickups).

In the initial phase, data primarily come from governmental reports (Ministry of Environment and Sustainable Development – Minambiente, National Planning Department – DNP, Secretariat of Economic Development of Bogotá) and recent academic studies. This approach has enabled the establishment of the project's foundations, provided strategic direction, and allowed for the evaluation of its viability and potential real-world impact on Bogota's society, particularly within the university community.

The analysis of international datasets—such as the one derived from the UNEP Food Waste Index Report 2021 (`Workspace/data/Food Waste data and research - by country.csv`)), covering 214 countries—further complemented and reinforced the project's viability.

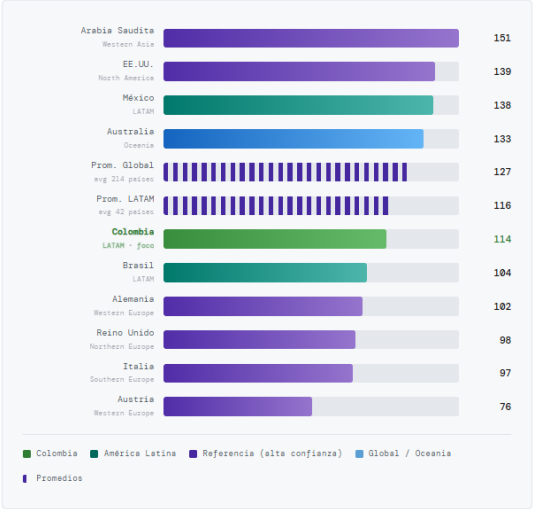

Fig. 1. Comparative chart of per capita annual food waste (in kg) across selected countries and regional/global averages, based on UNEP Food Waste Index Report 2021 (data via Kaggle).

The data indicate that Colombia wastes 114 kg per capita per year, below the global average (126.8 kg) and very close to the Latin American average (115.5 kg), although the total volume reaches an estimated 5.72 million tons annually. Notably, 61% of this waste occurs in households (70 kg per capita), representing the most challenging segment to intervene directly with a platform like the one proposed. Countries such as Austria (76 kg) and Italy (97 kg) demonstrate that significant reductions are achievable through effective policies, campaigns, and coordination tools. It is worth highlighting that Colombia's estimate has a very low confidence level according to the UNEP report itself, underscoring the scarcity of robust and updated national data on the phenomenon and reinforcing the need for localized initiatives such as this project.

In the context of the importance of reducing food waste, on October 4, 2024, the Mayor's Office of Bogotá launched the initiative "Yo Contribuyo, No Desperdicio" (I Contribute, I Do Not Waste), which emphasizes the relevance of coordinating efforts between the public sector, social organizations, and private companies to collectively and effectively address the issue.

III. System Analysis

A. Element and Relationship Analysis

To analyze the system in a structured manner, a decomposition into four main subsystems is performed, grouping the platform's operational functions:

1) User Interaction System
2) Surplus Management System
3) Assignment and Coordination System
4) Monitoring and Analysis System

1) User Interaction System: This subsystem manages all communication between users and the platform.

- **User Registration and Management:** Account creation, user authentication, profile management, and user classification.
- **Mobile Interface (Frontend):** Display of available surpluses, publication of surpluses by cafeterias, collection confirmation, and pickup map visualization.
- **Notification System:** New surplus alerts, assignment confirmations, pickup reminders, and collection status updates.

2) Surplus Management System: This subsystem handles the registration, storage, and tracking of food items for redistribution.

- **Surplus Registration:** Food item information entry, estimated available quantity, availability time, and collection deadline.
- **Surplus Geolocation:** Automatic capture of establishment location, map representation, and verification that the point is within the operating radius.
- **System Database:** User profile storage, surplus publication records, redistribution history, and impact metrics log.

3) Assignment and Coordination System: This subsystem is the operational core, responsible for coordinating food redistribution between suppliers and recipients.

- **Matching Algorithm:** Evaluation of geographic proximity, analysis of user availability, prioritization of nearby users, and selection of the best candidate to collect the surplus.
- **Assignment System:** Sending collection requests to the selected user, managing acceptance or rejection, and reassigning surpluses in case of rejection.
- **Collection Tracking:** Confirmation that the user collected the food, recording of redistribution completion, and surplus status updates.

4) Monitoring and Analysis System: This subsystem enables evaluation of the system's impact and performance.

- **Activity Log:** Record of all transactions, active user tracking, and redistribution history.
- **Administrative Dashboard:** Visualization of impact metrics, system activity analysis, and identification of usage trends.
- **Metrics Generation:** Calculation of recovered food, number of redistributions performed, collection success rate, and user participation.

B. Relationships

1) User Interaction Subsystem:

- **Mobile app ↔ User Registration and Management:** The mobile app allows users to create accounts, log in, and manage their profiles. The registration module validates and stores this information in the database.
- **Mobile app ↔ Notification System:** The app receives push notifications when new surpluses are available or when a collection is assigned to the user.
- **User Registration ↔ Notification System:** The notification system uses profile information (ID, device, preferences) to determine who receives alerts.

2) Surplus Management Subsystem:

- **Surplus Registration ↔ Database:** Each surplus published by a restaurant is stored in the database along with its description, quantity, and deadline.
- **Surplus Registration ↔ Geolocation:** When a surplus is published, the system automatically captures the establishment's location to display it on the map.
- **Geolocation ↔ Database:** Geographic coordinates are stored in the database so the matching algorithm can calculate distances between suppliers and users.

3) Assignment and Coordination Subsystem:

- **Matching Algorithm ↔ Database:** The algorithm queries the database to retrieve information on available surpluses and active users.
- **Matching Algorithm ↔ Assignment System:** The algorithm selects the most suitable user and forwards that decision to the assignment system.
- **Assignment System ↔ Notification System:** Once a candidate is chosen, the assignment system sends a notification to accept or reject the collection.
- **Assignment System ↔ Collection Tracking:** When the user accepts the request, the system activates the tracking module to monitor the collection.
- **Collection Tracking ↔ Database:** Upon completion of the collection, the transaction is recorded in the system history.

4) Monitoring and Analysis Subsystem:

- **Activity Log ↔ Database:** All system actions (publications, assignments, collections) are recorded in the database.
- **Metrics Generation ↔ Activity Log:** The metrics system analyzes activity records to calculate indicators such as number of redistributions or recovered food.
- **Administrative Dashboard ↔ Metrics Generation:** The dashboard visually displays the generated indicators to facilitate performance evaluation.

5) Cross-subsystem Relationships:

- **Mobile app ↔ Surplus Registration:** Restaurants use the app to publish food surpluses.
- **Mobile app ↔ Geolocation:** The app retrieves the device's location to register or display pickup points.
- **Mobile app ↔ Assignment System:** Allows users to accept or reject collection requests.
- **Mobile app ↔ Collection Tracking:** Users confirm food collection directly from the app.
- **User Registration ↔ Matching Algorithm:** The algorithm uses information on available users to perform assignments.
- **User Registration ↔ Activity Log:** User actions are recorded for subsequent analysis.

- **Surplus Registration ↔ Matching Algorithm:** Registered surpluses trigger the process of searching for nearby users.
- **Surplus Registration ↔ Activity Log:** Each surplus publication is recorded as a system event.
- **Geolocation ↔ Matching Algorithm:** The algorithm uses location data to calculate distances between users and surpluses.
- **Geolocation ↔ Assignment System:** Geographic proximity helps select the nearest user.

C. Sensitivity and Variability Study

This section analyzes the system's responses to critical input variations, identifying the most sensitive parameters, emergent behavioral patterns, and chaotic dynamics across the four subsystems.

1) Critical Input Sensitivities:

- **High-Demand Surplus Events:** Situations in which a surplus posting attracts a significantly higher number of claim requests than available portions, generating contention among users. The matching algorithm must resolve concurrent requests under time pressure, increasing the risk of assignment conflicts, duplicate notifications, or user dissatisfaction. The assignment logic must enforce strict request serialization to prevent race conditions and ensure that only one user is assigned per available surplus unit. *Affected subsystems: Assignment and Coordination System, User Interaction System.*
- **Large Volumes of Simultaneous Users:** A large number of students or community members attempting to access, browse, or claim surplus food at the same time, particularly during peak meal hours. Concurrent load on the mobile interface and database may degrade response times, delay notifications, or cause inconsistent availability displays across devices. Backend scalability and real-time database synchronization are essential to maintain system integrity under high concurrency scenarios. *Affected subsystems: User Interaction System, Surplus Management System.*
- **Geolocation Service Failure:** Unavailability or degraded performance of the external geolocation provider, preventing accurate capture of user or establishment coordinates. Without valid location data, the matching algorithm cannot calculate distances, surplus listings cannot be mapped, and the assignment process is effectively halted. As an external dependency, geolocation services represent a single point of failure; the system should implement fallback mechanisms such as manual location input or cached last-known coordinates to maintain partial operability during outages. *Affected subsystems: Surplus Management System, Assignment and Coordination System.*
- **Confirmed Match with No Pickup:** A user accepts a surplus collection request but does not complete the pickup within the defined time window, leaving the food unclaimed and at risk of being wasted. The surplus remains blocked for other potential recipients during the acceptance window, reducing redistribution efficiency. The system must implement automatic reassignment triggers upon timeout expiration, combined with a user reliability scoring mechanism to deprioritize or restrict accounts with recurring non-completion patterns. *Affected subsystems: Assignment and Coordination System, Monitoring and Analysis System.*

2) Emergent Behaviors:

- **Informal Priority Networks Among Users:** Over time, frequent users may develop informal coordination strategies outside the platform — such as messaging groups or agreed-upon pickup schedules — to secure surplus food before it is assigned through the algorithm. These external coordination patterns can distort the matching algorithm's assumptions about user availability and demand distribution, reducing the fairness and efficiency of automated assignments. The system should monitor anomalous acceptance patterns and consider incorporating behavioral signals into the matching logic to detect and mitigate informal workarounds. *Affected subsystems: User Interaction System, Assignment and Coordination System.*
- **Supplier Disengagement Under Low Demand:** If surplus postings consistently go unclaimed — due to low user activity, unfavorable pickup windows, or poor location coverage — cafeterias and restaurants may gradually reduce their participation in the platform. A reduction in supplier activity decreases available food listings, which in turn reduces user engagement, potentially triggering a self-reinforcing disengagement cycle on both sides of the platform. The administrative dashboard must surface early warning indicators of declining supplier activity so that operational interventions can be made before disengagement becomes structural. *Affected subsystems: Surplus Management System, Monitoring and Analysis System.*
- **Inequitable Distribution Patterns:** The geographic proximity bias inherent in the matching algorithm may systematically favor users located closest to supplier establishments, leaving users at the periphery of the operational radius consistently underserved. Over time, peripheral users may perceive the platform as inaccessible and disengage, concentrating active participation in a small geographic cluster and reducing the platform's broader social impact. Periodic equity audits of assignment distributions should be conducted, and the algorithm should incorporate distance-balancing mechanisms to prevent chronic geographic exclusion. *Affected subsystem: Assignment and Coordination System.*

3) Chaotic System Dynamics:

- **Nonlinear Cascade from a Single No-Show:** A single uncompleted pickup can initiate a cascade of reassignments, each with its own acceptance window, progres-

sively consuming available time until the surplus expires unclaimed. What begins as an isolated behavioral failure propagates through the assignment pipeline in a nonlinear fashion, with the probability of successful redistribution decreasing sharply with each reassignment cycle. Reassignment timeout windows must be dynamically adjusted based on remaining surplus viability time, compressing decision windows as the expiration deadline approaches. *Affected subsystems: Assignment and Coordination System, Surplus Management System.*

- **Demand Amplification Through Notifications:** When the notification system simultaneously alerts a large number of users about a newly published surplus, it can generate a demand spike disproportionate to the available quantity, causing system congestion and user frustration. The sudden convergence of simultaneous claim attempts on a single posting overwhelms the assignment logic and degrades the user experience for the majority of respondents. Notification delivery should be staggered or segmented by proximity tier to distribute demand more evenly across time. *Affected subsystems: User Interaction System, Assignment and Coordination System.*
- **Feedback Loop Between Metrics and Supplier Behavior:** If the administrative dashboard prominently displays low redistribution rates or high no-show percentages, suppliers may interpret these metrics as signals of platform unreliability and reduce their posting frequency. The metrics intended to improve system transparency can inadvertently function as a negative feedback signal, reinforcing supplier withdrawal and further degrading the indicators they were meant to improve. Metric presentation in the dashboard should be contextualized with trend data and benchmarks, and suppliers should receive individualized performance summaries rather than system-wide aggregate statistics. *Affected subsystems: Monitoring and Analysis System, Surplus Management System.*

D. Finding Patterns and Recognition

The platform generates operational and behavioral data through user interactions, food listings, reservations, and pickup confirmations. Analyzing this data allows the identification of relevant patterns that can improve the efficiency and impact of the food redistribution system.

1) Data-Driven Insights: The system collects data related to food availability, reservation frequency, pickup completion rates, and user activity. By analyzing these datasets, the platform can extract insights such as peak hours of food availability, the types of food most frequently redistributed, and the average time required for surplus items to be claimed. These insights allow administrators to better understand the dynamics of supply and demand within the platform and adjust operational strategies accordingly.

2) Trend Analysis: Trend analysis focuses on identifying recurring behaviors over time. The system may observe patterns in surplus food generation at specific times of the day, such as after lunch service or near closing hours. Similarly, user demand may follow predictable trends, such as increased activity during weekdays or academic periods. Recognizing these trends allows the platform to anticipate supply-demand fluctuations and improve coordination between suppliers and recipients.

3) Anomaly Identification: Anomalies represent unusual or unexpected system behaviors that may indicate operational problems or misuse of the platform. Examples include:

- Sudden spikes in unclaimed food listings.
- Unusually high reservation cancellations.
- Repeated no-show pickups by certain users.

Detecting such anomalies helps administrators intervene early, adjust policies, or provide additional support to maintain system efficiency and reliability.

4) Behavioral Patterns: User behavior analysis helps identify how different stakeholders interact with the system:

- **Recipients:** Some users may consistently claim food immediately after it is posted, while others interact with the platform only occasionally.
- **Suppliers:** Cafeterias and restaurants may exhibit patterns in how often they publish surplus food or the quantities they provide.

Understanding these behaviors allows the platform to improve notification systems, recommendation algorithms, and user engagement strategies.

E. Critical Assessment

Evaluating the performance of the Food Waste Reduction Platform is essential to ensure that it effectively fulfills its objective of reducing food waste while maintaining reliable and efficient operations.

1) System Performance Evaluation: System performance can be assessed through the following operational metrics:

- Response time of the mobile application.
- System uptime and availability.
- Successful reservation rates.
- Proportion of surplus food successfully redistributed.

Monitoring these indicators allows administrators to determine whether the platform operates efficiently and meets the needs of both suppliers and recipients.

2) Bottleneck Identification: Operational and technical bottlenecks may arise in several areas of the system:

- **Operational:** Limited pickup windows may prevent users from claiming available food, while delays in listing publication may reduce the chances of successful redistribution.
- **Technical:** Slow server response times during peak usage periods or insufficient notification delivery may prevent users from reacting quickly to new listings.

Identifying these bottlenecks highlights areas where targeted improvements are necessary.

3) Optimization Opportunities: Several optimization opportunities can improve system performance:

- Improving notification systems to alert users in real time when new food becomes available.

- Implementing recommendation algorithms that prioritize nearby users.
- Adjusting pickup windows to increase accessibility for a broader user base.
- Improving supplier interfaces to encourage more consistent participation from food providers.

4) Recommendations: Based on the analysis, the following recommendations are proposed to improve the effectiveness of the platform:

- Implement real-time notification mechanisms to reduce the time between food publication and reservation.
- Introduce predictive analytics to anticipate surplus generation and match it with potential demand.
- Develop monitoring dashboards for administrators to track operational metrics and detect anomalies.
- Encourage supplier participation through simplified listing interfaces and automated surplus reporting.
- Establish policies to manage repeated reservation cancellations or no-show pickups.

These measures would improve system efficiency, increase the success rate of food redistribution, and enhance the overall impact of the platform.

IV. System Diagrams

This section presents the main diagrams used to describe the structure and relationships within the proposed system.

A. System Component Diagram

Figure 2 presents the main components of the system and how they interact with each other.

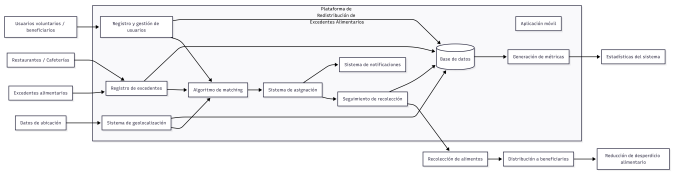

Fig. 2. Component diagram of the proposed system showing the main modules and their interactions.

B. System Relationship Diagram

Figure 3 illustrates the relationships between the different elements of the system.

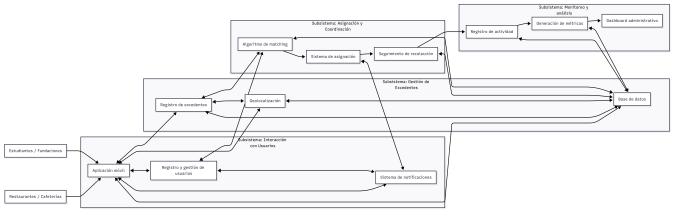

Fig. 3. Relationship diagram representing the interactions between system elements.

V. Conclusions

This work analyzed the design and potential impact of a digital platform aimed at reducing food waste through the redistribution of surplus food from university cafeterias and local restaurants in Bogotá. By leveraging mobile technologies, geolocation, and real-time coordination, the system enables more efficient connections between food donors and recipients.

The study highlights how technological solutions can support sustainability efforts by minimizing food waste while improving access to available food resources within the community. Additionally, the analysis identified opportunities for system optimization, including improved monitoring, real-time notifications, and data-driven decision support.

Overall, the proposed platform demonstrates the potential of digital tools to address food waste challenges and promote more sustainable and collaborative resource management.